

A Magical Walks in a Magic Forest (magicforest)

In a magical forest far-far away lives the Miraculous Deer¹. It likes to take *magical walks* along the trails of the forest. You're wondering, how many distinct *magical walks* can it take?



Figure 1: The deer on the way.

The forest contains N meadows, numbered from 1 to N . There are M trails, numbered from 1 to M . Trail i connects meadows a_i and b_i , and has a magical value c_i . A walk starts at a meadow and visits a number of meadows by traversing trails, arriving at a meadow. The walk might visit the same meadow multiple times. The length of the walk is the number of traversed trails.

A walk of length k is a *magical walk*, if the magical values of the traversed trails in order are $m_1, m_2, \dots, m_k$, and

- $k \geq 1$,
- $m_i + 1 = m_{i+1}$ for all $1 \leq i \leq k - 1$.

Two *magical walks* are different if the sequences of traversed trails are different.

Write a program that calculates the number of different *magical walks* modulo $10^9 + 7$!

 Among the attachments of this task you may find a template file `magicforest.*` with a sample incomplete implementation.

Input

The first line contains the integers N and M .

Each of the following M lines contains three integers a_i , b_i and c_i , representing a trail connecting meadows a_i and b_i with a magical value c_i .

Output

You need to write a single line with an integer: the number of *magical walks* modulo $10^9 + 7$.

 The *modulo* operation ($a \bmod m$) can be written in C/C++/Python as `(a % m)` and in Pascal as `(a mod m)`. To avoid the *integer overflow* error, remember to reduce all partial results through the modulus, and not just the final result!

¹Hungarian mythological creature

Notice that if $x < 10^9 + 7$, then $2x$ fits into a C/C++ `int` and Pascal `longint`.

Constraints

- $2 \leq N \leq 500\,000$.
- $1 \leq M \leq \min\left(\frac{N(N-1)}{2}, 1\,000\,000\right)$.
- $1 \leq a_i \neq b_i \leq N$ for each $i = 1 \dots M$.
- $1 \leq c_i \leq 10^9$ for each $i = 1 \dots M$.
- Any pair of meadows is connected by at most one trail.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (17 points) $M = N - 1$ and $a_i = i, b_i = i + 1$ for each $i = 1 \dots N - 1$.

- **Subtask 3** (19 points) $c_i \leq 3$ for each $i = 1 \dots M$.

- **Subtask 4** (14 points) $N \leq 22$ and $M \leq 22$.

- **Subtask 5** (20 points) $N \leq 1000$ and $M \leq 5000$.

- **Subtask 6** (30 points) No additional limitations.


Examples

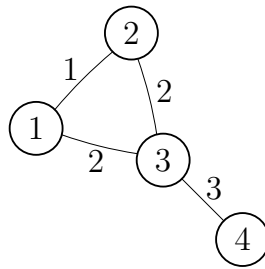
input	output
4 4 1 2 1 2 3 2 3 4 3 1 3 2	10
4 3 1 3 3 3 4 2 3 2 1	5
3 3 1 2 1 2 3 2 3 1 3	6

Explanation

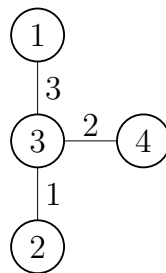
In the **first sample case** there are 10 *magical walks*:

- length of 1: $1 - 2$, $2 - 3$, $3 - 1$, $3 - 4$,
- length of 2: $1 - 2 - 3$, $2 - 1 - 3$, $2 - 3 - 4$, $1 - 3 - 4$,
- length of 3: $1 - 2 - 3 - 4$, $2 - 1 - 3 - 4$.

Note that, by definition, the magical walk $2 - 1$ is not different from the magical walk $1 - 2$, as the sequences of traversed trails are not different for these two walks. Therefore, the magical walk $2 - 1$ was not taken into account.



In the **second sample case** there are 3 *magical walks* of length 1, and 2 walks of length 2.



In the **third sample case** there are 3 *magical walks* of length 1, and 2 walks of length 2, and 1 walk of length 3.

